

CHAPTER 10

Influence of blockchain technology in pharmaceutical industries

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1. Introduction

Blockchain technology is a distributed, secured log file or shared ledger with a mechanism to trust all the transactions without any centralized authority (third party). Blockchains are primarily expedited on a platform of databases with some inherent pre-agreed technical and business logical criteria, using peer-to-peer mechanisms and pre-agreed consensus algorithms. Immutable data are being stored in blockchains, meaning that the data or objects cannot be modified after they are created. A few important points about the blockchain technology are described next:

- A shared ledger mechanism permits any participant within the business network to view the established ledger or a record (via distributed consensus).
- All the viable group activities are maintained in the shared public ledger.
- Every peer address is anonymous, and multiple addresses could be assigned to the same transactor.
- All the transactions are placed in blocks, which are connected by a one-way hash function.
- This will work in peer-to-peer mode, based on the Domain Name System and seed nodes.

Electronic health records, drug supply, education, biomedical research, and health insurance are some of the applications in the medical domain. The utilization of this blockchain technology is still immature, with lack of professional expertise, which makes it difficult to have a strong strategic perception to identify its potential. Still, there are several challenging issues with scalability, user adoption, and securing smart contracts. Health

practitioners should be aware of using this blockchain technology to achieve more benefits. This chapter describes the role of blockchain technology in the pharmaceutical industry by covering the benefits and challenges of incorporating this technique.

1.1 Background study of blockchain

The objective of blockchain is to handle digital content by storing and distributing it as received. This technology was developed by two scholars, Stuart Haber and W. Scott Stornetta. They had a plan for the deployment of a secure system to ensure that the timestamps in a document cannot be altered. Almost 2 decades later, the first blockchain was devised by Satoshi Nakamoto, which functioned as a public ledger. This was mainly used for enabling transactions for Bitcoin in 2009. Nakamoto constructed Bitcoin as a form of money that could be passed among peer-to-peer nodes without a bank or other centralized authorities, where a ledger is maintained (Muniandy et al., 2019). Bitcoin uses the technology stack of blockchain to secure the ledger payment record, and it also achieves the immutable property of the record (Nakamoto, 2008). The evolution of blockchain is classified into three phases:

(i) Transaction phase: blockchain 2.0 and Bitcoin emergence (2008–13)

Bitcoin, which hit the airwaves in the form of packages, has cropped out in such a way to leverage the concepts and abilities of the virtual ledger technology. Consequently, blockchain records incorporate a prolonged listing of packages that evolve rapidly.

(ii) Contract phase: blockchain 2.0 and Ethereum development (2013–15)

Vitalik Buterin commenced a malleable blockchain to introduce various distinct features in a peer-to-peer network. Ethereum was recognized as a modern-day public blockchain in 2013 with different functionalities in comparison with Bitcoin, an improvement that emerged in blockchain history.

(iii) Application phase: blockchain 3.0 and the future (2018)

Blockchain technology is the biggest innovation of the 21st century and is widely used in various sectors like cryptocurrency exchange, finance and banking, supply chain, overall resource management and tracking, voting mechanisms, publicizing sectors, construction of unique and innovative content, and the educational sector (Astarita et al., 2020). For example, banking institutions are established to connect different groups of people together and allow all kinds of commercial activities and trade between

them with centralized controlling authorities. Blockchain also accomplishes the same except it is decentralized and no single entity can own it (Amin, 2020). Figs. 10.1 and 10.2 show banking transactions with and without using blockchain technology.

Blockchain can make trade efficient by removing the manual paper-based process and introducing a trusted, streamlined, automated process. A few benefits are given next:

- faster payment processing
- simplified digital identity verification
- effective peer-to-peer transactions
- hedge fund management
- buying and selling assets
- fund raising, credit, and loans

The need for blockchain technology is described subsequently.

Decentralized: Customers participate in the blockchain network using their devices, and it does not require any governing authority. The participation is granted immediately to carry out further activities.

Immutability: Immutability describes how the blockchain ledger stays unchanged, intact, and inefaceable. Moreover, records inside the

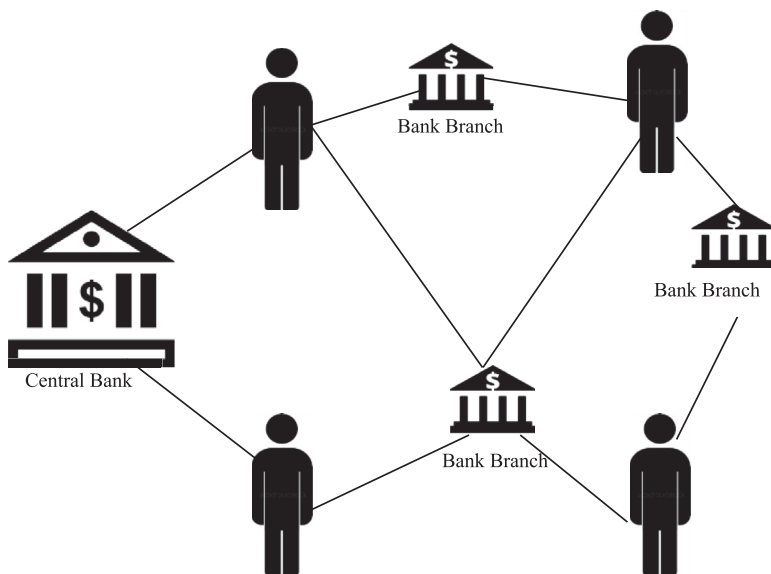


Figure 10.1 Current banking system.

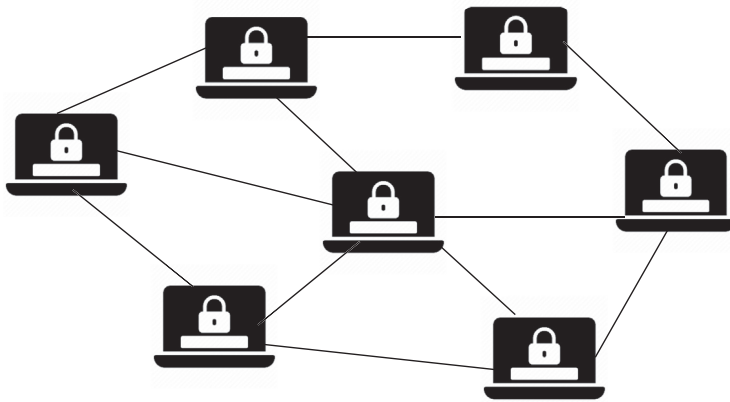


Figure 10.2 Blockchain-based banking system.

blockchain cannot be altered. Each block of information, which includes records or transaction details, continues the use of a cryptographic precept or a hash value.

Security: Information in the blockchain is hashed cryptographically. All the blocks inside the ledger include a unique hash code and the hash code of the preceding block. It will offer a specific identity for each piece of information. It is very hard to modify or tamper with the information, which suggests converting the complete hash IDs, and the hash value is also irreversible, making it more robust.

Distributed ledgers: A public ledger maintains the information regarding the transaction and the participant. These facts are known to the customers because the distributed ledger is maintained among the users who are in the community. This mechanism affords robustness throughout the complete process, which assures a better outcome (Pandey and Litoriya, 2021; Hyperledger).

Consensus: Consensus algorithms highly support the decision-making procedure among the participants. In a distributed environment, a consensus algorithm is a technique that enable all parties in a blockchain network to agree on the shared data state of the distributed ledger and trust unknown peers. Nodes now no longer believe every other node; however, they can believe the consensus algorithms that are executing on them.

Faster processing: Traditional banking structures are less powerful compared with using blockchain technology. This can permit quicker settlements for any type of contract, and it facilitates the consumer to switch cash rapidly, which reduces time consumption (Azzi et al., 2019).

Some of the challenges in blockchain technology are described next.

Complexity of blockchain: Blockchain is constructed using cryptographic mechanisms to offer protection and set up consensus over a disbursed network. Some complicated algorithms have to be executed to authenticate users, which in turn introduces computational overhead. Furthermore, storing a worldwide blockchain with “m” people containing “n” precise blocks as a set of lists, with the worst-case state of affairs, results in $O(m.n)$ memory (Viriyasitavat and Hoonsopon, 2019).

Scalability: It is a primary problem in the overall performance of the system, even with fault-tolerant systems using authorization-based blockchains. The number of transaction nodes is not restricted, and it can be used for large-scale applications, but it will slow down the processing of transactions.

51% attack: In a public blockchain environment, a 51% attack is a malicious miner or a collection of miners taking advantage of greater than 50% of a network’s mining strength or hash rate.

Operations are sluggish and cumbersome: Due to the consensus mechanism and extra layers of encryption, the procedure is slower compared with conventional processing. This postponement creates a primary risk in the public blockchain.

Energy consumption: Energy is consumed to perform the complicated mathematical computations to trigger and verify the procedure, which makes the complete network stable. In general, Bitcoin core takes up around 200 GB storage area in each node within the blockchain network. In addition, it requires 5 GB of storage to add and 500 MB to download every day.

The blockchain consists of application, distributed computing, platform, and infrastructure layers, and it is depicted in Fig. 10.3. Each block has a block index, timestamp, previous hash value, current hash value, and data. Each block is secured with a hash code, which is irreversible. The block details are shown in Fig. 10.4.

Ever since Satoshi Nakamoto released Bitcoin in 2009, cryptocurrency has reached our footsteps. More recently, people are using blockchain for various real-time applications such as real-time IoT, medical data sharing, digital IDs, digital voting, cloud storage, real estate marketing, advertising insights, and supply chain and logistics monitoring (Hyperledger).

1.2 Blockchain in the pharmaceutical and healthcare industries

The advantage of blockchain technology is its capability to create verifiable trials and launch drug sourcing throughout the supply chain. Pistoia

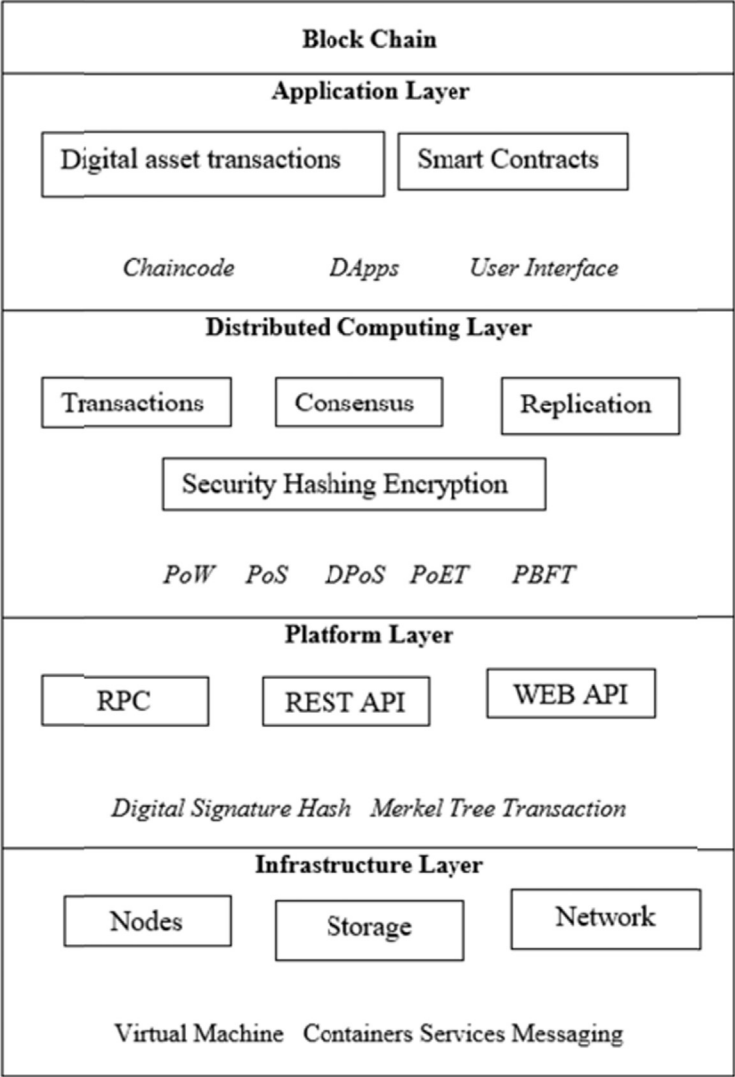


Figure 10.3 Layered architecture of blockchain.

Alliance survey says that around 70% of pharmaceutical and life science leaders trust that blockchain will create a larger impact in the pharmaceutical and healthcare industries. Decentralized blockchain solutions allow manufacturers and their customers to autonomously validate the quality and origin of drugs rapidly and securely (Chen et al., 2018). Transparency and security are the two important factors of blockchain technology for the

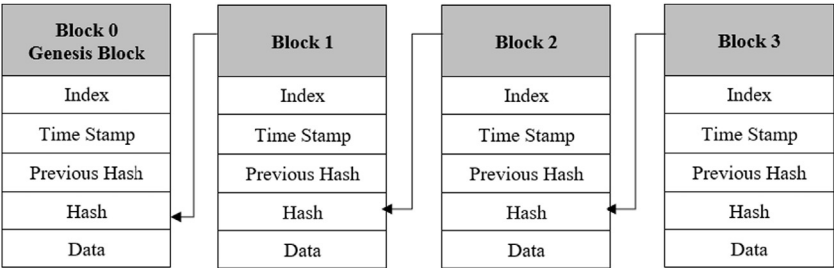


Figure 10.4 Block details.

users in pharmaceutical and healthcare industries. All stakeholders in the supply chain need to share and update data and ensure it is on time. Using this technology, the entire supply chain can be managed with a single software shared among legitimate stakeholders. Pharmaceutical companies and their suppliers, payers, pharmacies, and patients can access the real-time data. With this technology, a single piece of software is shared across the supply chain among legitimate stakeholders. The industry people and their stakeholders can access the data when it is released. Blockchain technology can also help in avoiding diversion, fabricating, and interfering. Medicines for the patients can be traced from manufacturer's end to reaching the market. If any attempt is made in modifying the registry, it would be immediately visible to all involved parties.

1.2.1 Patient data management

Blockchain technology is expected to simplify the process of managing patient data between service providers, including different insurance companies and R&D institutes, using a unique identifier. The data is gathered from portable devices, always where the patient is, including what tests they have undergone, and can be used in other transactions. Perhaps most importantly, the patient can regulate the granting of access. This system considerably reduces the management costs. This requires great cooperation among the parties. A scalable system with standard approval and authentication capabilities could be deployed with limited real-world patient data. The patient management system could be implemented with primary medical records, and accessibility would be managed by the patient (Khatoon, 2020).

1.2.2 Payer data management

The pharmacy benefit manager acts as an intermediary between pharmacies, medicine companies, and paymasters by managing formulas, negotiating concessions and refunds, and processing and paying for drug orders.

Usually, an electronic platform is used, and it reduces waste, price fluctuations, and improves customer understanding. For example, blockchain technology and smart contracts can speed up the life insurance approval process and speed up the processing of claims. Distributed ledger technology (DLT) tracks bills throughout their lifecycle and tracks up to 50 million transactions per day for many patients. For small-scale industry, some insurance companies, providers, and testing service leaders are participating in the Synaptic Health Alliance. Focus is on preserving an exact provider directory. Moreover, it highly focuses on developing a physician license verification system and it is significantly required in healthcare and Medicaid service centers.

1.2.3 Blockchain infrastructure

Blockchain architecture is built by focusing on data sharing and transaction tracking throughout the pharmaceutical industry supply chain. The application-oriented software companies and many startups are focused on evolving solutions that can take advantage of blockchain technology. In addition to developing these physical systems, the acceptance of data exchange is also promoted. This requires participation and practice of the security and privacy features of a blockchain solution (Wu and Lin, 2019).

Meanwhile, several small efforts have started to test the potential of blockchain technology for the benefit of the pharmaceutical industry. A pharmaceutical company worked with SAP to build the SAP Pharma Blockchain POC (proof of concept) application for medicine tracking. Since its release in 2017, it has been repeated many times. In early 2019, SAP declared the readiness of the SAP Information Partnership Center for Life Sciences, a new solution designed to help customers comply with the US Pharmaceutical Supply Chain Security Act.

1.2.4 Problems faced in the pharmaceutical industry

Among the various healthcare applications, the most significant and crucial part is the pharmaceutical industry. The functioning of pharmaceuticals is done in three phases, namely, discovery, development, and finally the distribution process. Various issues that arise in the pharmaceutical industry are described next.

1.2.4.1 Data disparity in pharmaceuticals

It has been observed that this is the most conventional and common issue that has been faced by several pharmaceutical industries. Discrepancy in data

arises during the development of the drug. Normally, pharmaceutical data is recorded and placed where it needs a proper accessing methodology across the platforms. Departments that are associated with this business adopt an individual data structure and model. Accessing the sensitive data for further analytics needs more focus, and it is considered a crucial task. There is a necessity for the amalgamation of technologies like blockchain and big data analytics, so efficiency and accuracy can be enhanced (Rabah, 2017). Fig. 10.5 shows the challenges in the pharmaceutical industry.

1.2.4.2 Collection and data handling

Data that is used for analysis comes from diverse sources. Collection and managing the data is a time-consuming process since it might vary in its form. Apart from handling, analyzing the data is also a challenging task due to its nature.

1.2.4.3 Ambiguity and data inaccuracy

Ambiguity exists in the pharmaceutical data. As the data could be accessed through several channels, there is no guarantee for the data to be new and informative. Through the deployment of blockchain in this domain, focus could be more on the data while it is registered and stored for further process (Khan et al., 2020). The implementation of the blockchain enables data to be sorted based on relevance and to be unambiguous.

1.2.4.4 Supply chain

The most significant issue in the pharmaceutical industry is to retain the supply chain, and it is a complex task (Tseng et al., 2018). In the supply

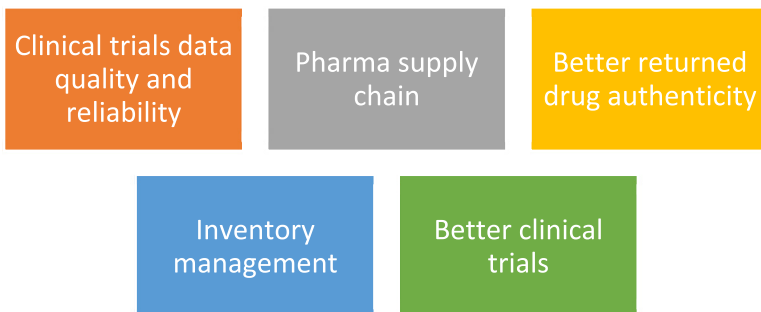


Figure 10.5 Solving pharmaceutical challenges using blockchain.

chain management system, it is found that there are more possibilities for great monetary loss, drug counterfeiting and fraudulent activities (Bryatov and Borodinov, 2019).

1.3 System architecture of blockchain in the pharmaceutical industry

Drug traceability is the process of accessing information that is associated with a drug during its life cycle, so evidences are recorded continuously. In this context, a paper discussed the utilization of the GS1 (General Specifications) standards barcodes that are used in Smart-Track (Al Huraimel and Jenkins, 2020). Bar codes are used to determine the identifier of the product, expiration date, and production number. This information is recorded during the process of several supply chains, so it ensures keeping a record of the transfers of ownership. There is a possibility of the verification of the authenticity by the end user since the repository is maintained globally. This is made possible with a smartphone app also. At the receiving end, the barcode on the product is used for the verification of its features. In Huang (Huang et al., 2018), a data matrix system is maintained for each drug. This denotes the various characteristics such as the identifier of the product, manufacturer identification, package ID, authentication code, and the metadata, which is optional. Through this information, it becomes easy for the patient to check the source of the drug (Lin et al., 2017). Complete transparency and visibility could be ensured with the help of near field communication technology in the supply chain.

Ethereum has been deployed by Figorilli (Figorilli et al., 2018) to perform anticounterfeiting. This solution works based on a smart contract. A drug ledger for a drug traceability system has been suggested (Huang et al., 2018). The proposed solution works similarly to the transaction logic of drugs that happens in the supply chain and thereby results in the privacy and authentication of the stakeholders' information. This work made use of the expanded UTXO (unspent transaction output) data structure. But the usage of UTXO data structure results in huge storage cost, less space utilization, and poor programmability (Wigand et al., 2011). A Hyperledger-based solution has been proposed toward drug traceability. Through this method, the throughput has been increased and latency is minimized. Moreover, the utilization of the resources is comparatively lower.

A detailed journey of a drug is created at each and every phase by scanning the barcodes. Later, these codes are stored on a ledger system that works based on the principle of blockchain. The inclusion of sensors in the

supply chain records the humidity, temperature, and other recorded parameters, and these data are stored in the ledger. The objective of the implantation of the sensors is to check the respective drugs that require refrigeration for storage (Caro et al., 2018). As soon as the arrival of the drug in a pharmacy or hospital, one can ascertain whether it has been compromised during its journey from the source. The audit trail reveals distinguishing features, and the checking process could also be recorded on the ledger. Through the implementation of the blockchain, it becomes easy for finding discrepancies. Certain benefits are as follows:

- Due to the drug traceability, the supply chain becomes easier.
- Various information such as expiration date leads to the control of the stock and rotation.
- Authentication and verification of drugs could be performed; thus, it results in the reduction of errors.
- Harm to the patient is reduced since the checking of drugs is enhanced.
- Drug counterfeiting is also reduced (Kumar and Tripathi, 2019).

1.3.1 Smart contract and blockchain tracking for handling drug counterfeiting

Various industries have deployed smart contracts and blockchain protocols for tracking the supply chain in the pharmaceutical industry (Molina et al., 2019). The inclusion of a DLT is also a good choice for handling counterfeiting. It has been observed in various articles that the adoption of the blockchain results in the reduction of fraudulent medical businesses. Fig. 10.6 will give the various application areas of the pharmaceutical industry (Hasan et al., 2019).

The automation of smart contracts is discussed next.

The nature of a smart contract is that it can safeguard funds. Adoption of Ethereum, Bitcoin, a stablecoin, or a Central Bank Digital Currency could also be used. In spite of practicing currencies, smart contracts will lead to several new supply chain practices:

- Escrowing of funds shall be done prior. The retailer makes use of the blockchain to pay in terms of digital funds through the smart contract. When the shipping company identifies the smart contract, then the funds will be released automatically (Lin et al., 2018).
- Shipping companies also utilize the concept of the smart contract so the payment shall be released. This results in huge benefits for the company.

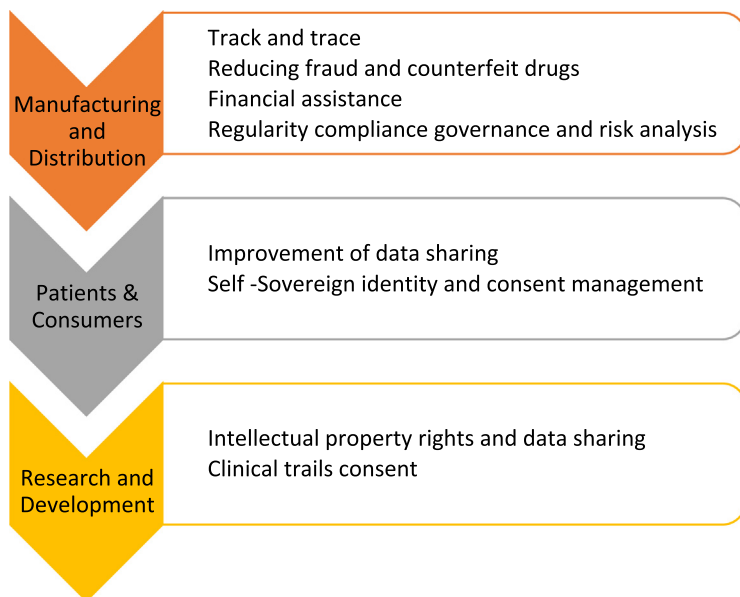


Figure 10.6 Blockchain in the pharmaceutical industry.

Apart from that, remittance done at the international level is found to be rapid and cost effective. Thus, it leads to a drastic improvement in the current financial system.

- Through the deployment of smart contract payments, the administrative costs are decreased and the efficiency is improved. During this process, the expense incurred in administration is greater since more attention is required in managing invoices and payments. Deployment of a smart contract is highly automated, so this results in the reduction of the supply chain costs (Xie et al., 2019).

2. Drug supply chain management in the smart pharmaceutical industry

The distributed and decentralized features of blockchain technology have prolonged the privacy and security of the pharmaceutical industry's drug supply chain. The full drug supply chain is presented as the blockchain network by involving participants where they can manage and update the activities of the entire supply chain. Blockchain-based systems store data relating to users of hospitals, pharmacies, manufacturers, distributors, suppliers, doctors, and patients. The major components in drug supply chain

management are drugs, order details, record repository, and raw material. The users in this system receive a client-side, application-based front end where their transaction can be quickly carried out and interact with the blockchain network (Williams and McKnight, 2014). By using the client application, the status of the drug delivery can be monitored by all the concerned participants during the entire process. A stored-off blockchain is a separate data library that acts like a data storage pool. It is an efficient tool used for data analytics and visualization since data visualization can be exploited in the medical-related institutes. In this, a machine learning model can endorse the best medicine to the patients. Furthermore, the participants belonging to this framework obtain permission to search the complete details of medicines, raw material stock, and further relevant information such as manufacture date, expiration date, price, etc.

To achieve the goal of distributed ledger consistency, actively participating peer nodes in the system are responsible for executing the consensus algorithm. The first supplier initiates the process of sending required raw materials to the drug company. With the web application portal, the participants involved in the system can be logged in where the transaction can be carried out by each user. For instance, if a manufacturer places orders for the raw material, every peer node in the blockchain network would be involved in validating the transaction; then the supplier receives the order. Once the order for the raw material is prepared, the supplier proceeds with the order confirmation. Doctors can place an order for medicines in a company using the blockchain system. Initially, a doctor should be authenticated. Proceeding with the transaction proposal submission process, the transaction is validated and confirmed. Then the status is confirmed with the doctor by communicating about the successful transaction (Yue et al., 2008).

The primary and significant purpose of the blockchain network is to accumulate data in a distributed manner, where several transactions are maintained in each block. Such transactions are maintained in an encoded form and hashed manner because of security issues. In the service-oriented architecture, the smart contract and ledger-oriented functionalities are offered as a service to users. End users such as suppliers, manufactures, dealers, pharmacies, clinicians, and patients will use the front-end web application to execute the transaction where they are able to carry out medicine orders, supply raw materials, update medicine details, update orders, and record verification, products supplied, data exchanged, drug monitoring, drug control, managing clients, etc. The primary objective of

the scheme is the prevention of counterfeit drugs, and this affords the end users with a secure scheme. With blockchain, each participant in the blockchain system can monitor the conveyance of the medicine.

In addition, it is capable of performing operations such as creating, reading, updating, and deleting among the connected peer nodes. Furthermore, to maintain security among the nodes, a private network can be introduced where data can be directly exchanged among nodes without revealing information to other nodes. For security purposes, every participant can build their own private network. Suppose the patients are not able to proceed with further transactions; they can check that the drug being bought from the pharmaceutical store is original or counterfeit by performing barcode scanning. Patients are provided details related to the drug manufacturing company such as manufacture date, expiration date, price, etc. This is possible because of this blockchain concept, which may also limit suppliers from just performing the raw material transaction with the manufacturer. And, the participants use the system as a private network.

The recommendation system functions to recommend the appropriate drug to pharmaceutical company customers. Sentiment analysis and machine learning techniques can be used on the data set, which contains comments provided by clients on the websites regarding whether the medicine is in good condition or not. The best medicine is recommended to the clients such as pharmacies, physicians, clinics, patients, and hospitals. With permissioned blockchain systems, unique participants are allowed to use this system. Moreover, users on the blockchain network are validated and registered by the administrator by supplying them with confidential credentials and registration certificates. Then, only the users are permitted to access the blockchain network.

In addition, it is the responsibility of the consensus algorithm to connect the registered private network users for executing and handling the transactions and the orders as well. In virtual environments, each participating node has a facility for exploiting smart contracts and a distributed ledger where they can maintain their data and execute transactions. Because of the smart contract, the nodes are capable of validating a transaction once after executing it and then write the transaction block into the distributed ledger. Furthermore, the distributed ledger holds all transaction histories, events, logging details of every transaction, and behavior carried out by the participants.

2.1 Smart contract in drug supply chain

The simple concept of a smart contract is to provide transparency for the parties to exchange properties, money, or shares without the intervention of an agent or broker of a third party. It is considered the best characteristic of blockchain networks. Smart contracts are technically viewed as multiple lines of computer code that implement the agreement among two parties without any extra payment to the third party. This is composed of a pre-defined set of rules on which the two parties agree. And, the smart contract is activated automatically once the stated condition satisfies the same requirements as the database events. In addition, intelligent contracts make it simpler for users to handle the privileges and their properties between various parties. Smart contracts are maintained in the blockchain platform's distributed ledger, where they are fully secured and protected against illegitimate activities such as deletion or tampering (Huang et al., 2018). Moreover, smart contracts are a safe method to execute business terms and conditions that could be implemented and prompted automatically by the blockchain network consensus algorithm.

Some significant technical challenges that arise with smart contracts are described as follows;

For writing a smart contract, a programming language called Solidity can be utilized. In this the learning rate is low and difficult to maintain. In addition, the rate of execution of all transactions is comparatively low and consumes excessive time to finish, since transactions are performed regularly or in sequence between all peer nodes of the network. This problem is solved by implementing smart contracts only for a specific set of nodes, which is not applicable for all network nodes. Healthcare supply chains are dynamic networks that cover numerous organizational and geographic borders, delivering vital services to daily life. Incorrect details, lack of clarity, and limited provenance of data may result in such systems. One result of such restrictions in the existing supply chains is counterfeit medicines, which not only have a severe adverse effect on human health but can also cause serious economic losses to healthcare industries. Thus, end-to-end monitoring systems for pharmaceutical supply chains have been highlighted in recent studies. To ensure product protection and eradicate counterfeits, an end-to-end monitoring system in the pharmaceutical supply chain is therefore imperative. Many recent tracking systems are centralized, leading to problems in healthcare supply chains with data protection, transparency, and authenticity.

An Ethereum-based blockchain system leverages smart contracts and automated off-chain storage for successful drug traceability in supply chain management. Furthermore, the smart contract assures data provenance, removes the necessity for any intermediary third parties, and offers all stakeholders a secure, immutable transaction background. To assess its efficacy and to increase traceability within pharmaceutical supply chains, testing and validation are performed along with cost and security analysis.

2.2 Transaction procedure in supply chain management

The procedure for executing transactions of drugs in the supply chain management is explained in this section. Initially, users are provided the interface of a client application that permits the user to join using their registered credentials for the blockchain system, and they can complete their requests for transactions. The registration of all users is the administrator's responsibility, and it is essential to carry out the transaction by a customer. To request a proposal for a transaction, initially the user logs in to the client application and exploits registered credentials to send his or her transaction request. After that, all the peer nodes can send the transaction proposal (Iakovou and Shi, 2019).

There are two groups of these peer nodes: committers and endorsers. The endorsers are responsible for enforcing or signing the transfer proposals and accepting the approval if it is legitimate or refusing it if it does not comply with the smart contract requirements. Committer peers are expected to confirm the outcomes of the transactions before loading them into the ledger's transaction block. Simply stated, endorser peers are the particular category of committer peers that hold the smart contract. In addition, in their own simulated environment, these endorser peers execute the smart contract of the demanded transactions before updating them into the distributed ledger while receiving proposals for the transactions.

In a simulated environment, known as the (read/write) RW package, the endorser peers would obtain the read and written data when the transactions are executed. The data that is in read mode contains the world state data, and in reality, written data is the data being written in the world state once the transaction is executed in a simulated environment. Then, the endorser peers with RW sets will transmit the signed transaction to the client. The customer sends the signed transaction to the consensus manager again with all RW sets, and the transaction is sent to the committer nodes, as well as the data being arranged into a block.

Afterward, by matching the present world state, the committer nodes verify the transaction, and then the transaction data is loaded into the distributed ledger. At last, according to the written records, the distributed ledger is updated. Ultimately, peer committers will send the transaction status update to the client either submitted or not submitted. Using REST API and a software development kit, communication between the client and the blockchain network occurs.

2.3 Distributed ledger in supply chain management

In business, the supply chain is a movement of knowledge and products by various suppliers, such as suppliers of raw materials, producers, distributors, retailers, end users, etc. SCM assists multiple users in the seamless tracking of products and data. The current supply chain management systems, however, lack information in transparency, user privacy, on-demand reports, inappropriate information monitoring, quality management, repudiation, user confidence, etc. An architecture for pharmaceutical supply chain management that includes an Ethereum blockchain, with DLT, can be used to solve these problems. This system helps to achieve consumer privacy, drug supply accountability, drug monitoring, quality control, and non-repudiation with increased confidence among consumers.

3. Blockchain technology to prevent counterfeit drugs

3.1 The problem of counterfeit drugs

At present, the problem of counterfeit drugs is one of the serious problems for the pharmaceutical industry and has created an increasingly acute issue for human health worldwide. The World Health Organization (WHO) reported that nearly 30% of the total drugs sold in developing countries are fake or counterfeit. According to WHO, counterfeit drugs are defined as “drugs that are manufactured fraudulently, low quality, mislabeled, hiding the source detail or identity and do not follow the defined standard” (Al-Jaroodi and Mohamed, 2019). While some counterfeit drugs may contain active ingredients, an improper amount of ingredients or impure production that may contain toxic ingredients can cause serious health problems.

Drug counterfeiting is one of the most lucrative businesses worldwide. One of the main reasons for counterfeit drugs is the pharmaceutical supply chain. Due to advancement in technology, production and distribution of counterfeited drugs has increased, and these drugs can be produced in large

quantities in a short amount of time. Due to lack of transparent drug delivery and supply chain process, it is difficult to detect counterfeited drugs. There are number of major gaps in pharmaceutical supply chains:

- fragmented and nontransparent supply chain infrastructure;
- number of intermediaries, making it difficult to trace the counterfeited drug's origin;
- adoption of different systems among the different drug companies that leads to multiple; solutions;
- lack of point-to-point systems infrastructure to keep data in sync across the healthcare supply chain.

To prevent counterfeit drugs, pharmaceutical companies need a secure and transparent drug supply chain management that can trace and track drug supply chains at every phase of the supply chain process, which includes raw material suppliers, drug manufacturing, distribution of drugs, pharmacies, hospitals, and consumers.

3.2 Blockchain: the solution

Blockchain is one of the latest innovations in the field of computing that can be considered an efficient solution for the problem of counterfeit drugs. The blockchain is a perfect solution where data security and privacy protection are major concerns in developing a secure drug supply chain. Blockchain is a decentralized distributed digital system that consists of a chain of blocks containing transaction data that can be shared across a network among the participants connected to the network in an encrypted form. All transactions are stored in a secure ledger with a timestamp and eliminate the need for third parties. A transaction, verified by the network participants using a majority consensus agreement protocol, can be added to blockchain. Any recorded transaction in a ledger cannot be modified or erased and can also be viewed at any point of time. To protect blocks in chains, hashing and digital signature techniques are used. There are a number of reasons to use blockchain technology in the pharmaceutical drug supply chain:

- **Extended security:** Blockchain is designed to provide 21st-century cybersecurity to prevent data and transactions from any illegal or unauthorized access. Using blockchain, trust can be increased among the participants in traditional drug supply chain systems, where they can anonymously exchange their transactions without third party interference.

- **Traceability:** Blockchain can also be used to track transactions on the network at any point of time. Therefore, blockchain is a suitable option to record the drug status throughout the entire drug supply chain process.
- **Transparency:** Blockchain maintains all recorded transaction history in a ledger, which helps to find the actual origin of the transaction in the network. This characteristic of blockchain brings more transparency to the drug supply chain where every drug changes ownership and new transactions take place.
- **Privacy protection:** Blockchain also provides privacy protection where every participant on the network is protected using a key-pair identifier. A transaction can be made from the public key of one participant to the public key of another. Using blockchain in the pharmaceutical supply chain, a patient's private medical data can be kept secret, and medical records can be used anonymously using a public key.

3.3 Drug supply chain

A drug supply chain in the pharmaceutical industry can be defined as a process where prescribed drugs or medicines are delivered to patients from manufactures (Poongodi et al., 2020). The basic entities in the drug chain are supplier, manufacturer, wholesaler distributor, packager, pharmacy, hospital, and also patients. The supplier supplies the raw material or ingredients for drugs sourced from different places to the manufacturer. The manufacturer ensures their inventory of drugs and sends the drugs to the distributor or wholesaler. The wholesaler or distributor delivers drugs to thousands of hospitals and pharmacies using transport agencies. The wholesaler can be categorized in two categories: primary or secondary. Primary wholesalers directly purchase drugs from manufacturers, whereas secondary wholesalers purchase drugs from other parties. Depending on the demand for certain drugs, a primary wholesaler may also purchase drugs from secondary wholesalers. The buying and selling of drugs or medicines between wholesalers and repeatedly repackaging of drugs by wholesaler is very common within the pharmaceutical industry. At last, pharmacies and hospitals sell drugs to the patient or consumer. Fig. 10.7 shows a typical drug supply chain scenario in the pharmaceutical industry.

In the drug supply chain, the drug moves among various entities, and every movement gives the ability for counterfeit drugs to invade the drug

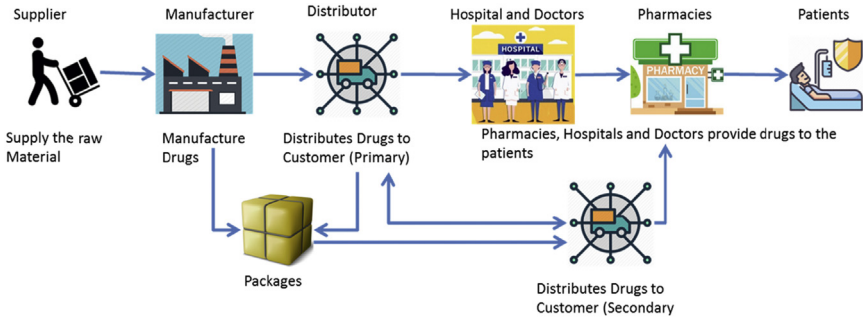


Figure 10.7 A typical drug supply chain.

supply chain. For example, fake drugs or medicines can be merged and be mistaken for legitimate drugs, or at the time of repackaging, false drugs might be given genuine labels at the wholesalers.

Therefore, drug traceability has become an important issue for public health officials, pharmaceutical industries, and government regulators. A reliable drug traceability system can track or follow the drug movement and benefit all stakeholders of the drug supply chain. Blockchain is a promising technology that can bring several useful changes to the drug traceability system in the drug supply chain. Blockchain-based systems could forbid data modification, ensure the authenticity, and enhance the security, flexibility, and durability of the system in peer-to-peer architecture.

3.4 Designing Drugledger

Drugledger is a framework for drug traceability and regulation that incorporates blockchain with drug supply chain (Thangamuthu et al., 2020). In the design of Drugledger, the following three principles are followed:

- (i) An expanded version of UTXO-based transaction model is used to construct the whole, true, practical drug supply chain workflow, especially for drug packaging, repackaging, unpackaging, drug order cancellation, drug manufacturing, and drug arrival.
- (ii) In the design of Drugledger, the service provider is separated into three independent roles:
 - **Certificate service provider (CSP):** A public key infrastructure service provided by the CSP is integrated into Drugledger that allows node participation in Drugledger with only a valid certificate. This property could be utilized to stop Sybil attacks where troublesome nodes could flood the network with several fake names and numbers of pointless or malicious transactions. Since in the

Drugledger the stakeholder's public key is recorded in the certificate, illegal participation of stakeholders could also be banned. Furthermore, the main role of every pharmaceutical stakeholder is also verified and added to the certificate, so Drugledger can deal with fake and meaningless drug production transactions. The CSP also ensures protection of stakeholder identity from the those that are not participating in transactions. The CSP can be the drug supply chain regulator and can be integrated with Drugledger.

- **Query service provider:** Drugledger allows each node in the network to act as a query service provider and enables participants to be free to register and cooperate with any number of query service providers of their choice. Then service providers can correlate participants' public keys with their physical identities in real life, for example, a pharmacy name or distributor name. Whenever a drug transaction happens in the Drugledger network, participants in Drugledger are needed to compute an appropriate one-way cryptographic hash function of the encoded raw information of the drug (barcode or the RFID label attached to drug package) as metadata. This encoded information will be treated as a future index for querying the service provider for tracking the drug information relating to a particular stakeholder. This information will be recorded in the blockchain-based Drugledger as a blockchain transaction across every replica.
 - **Antiattack service provider (ASP):** In Drugledger design, the ASP provides supervision for unusual and malicious activities such as abused transactions by suspicious stakeholders in the network to ensure smooth functioning. The ASP works as an anti-virus software for a Drugledger system that helps to detect possible attacks on the Drugledger network. [Fig. 10.8](#) shows the Drugledger basic workflow and service architecture.
- (iii) In Drugledger design, a scenario-driven, storage-pruning approach is used to solve the problem of continuously increasing storage in the blockchain system. In a real-world scenario of the drug supply chain, the workflow can be optimized by the expiration date of drugs. In Drugledger, the blockchain is pruned based on drug expiration date and executes the corresponding storage-pruning algorithm, so the blockchain can accomplish a stable storage.

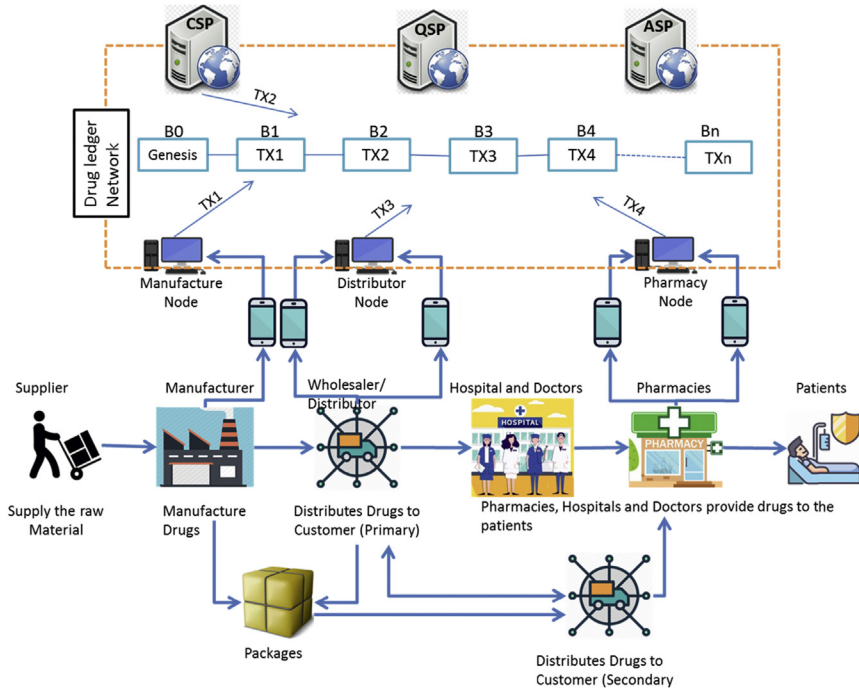


Figure 10.8 Drugledger basic workflow and service architecture.

3.5 Performance measures of Drugledger

- (i) Practicality, the first and important performance measure of Drugledger is its practicality in that it can be used to track the drug supply chain in a real scenario. Following are the key concepts that are used in the design of Drugledger to trace and regulate the practical drug supply chain in the pharmaceutical industry.
- (ii) Drugledger reconstructs the service provider architecture by dividing it into three different service layers to ensure a practical, sustainable service delivery system and also maintains data privacy protection and authentication.
- (iii) As a use of blockchain in the drug supply chain, it records drug transaction data to trace drug status throughout the drug supply chain from manufacturers to patients.
- (iv) Drugledger uses a scenario-oriented storage prune approach to decrease blockchain storage with drug expiration date, which enables blockchain to achieve stable and acceptable storage.

- **Security:** The blockchain system is permissioned to counter Sybil attacks in Drugledger design. The CSP and ASP limit the access control of participants to Drugledger by prohibiting network access among commercially ill-reputed or suspicious stakeholders in the network. Due to peer-to-peer architecture, Drugledger is also less prone to denial-of-service attacks in comparison to traditional client-server architectures.
- **Efficiency:** The efficiency of Drugledger is collectively determined by the blockchain protocols and service providers' communication overhead.
- **Transparency:** As blockchain records all drug transaction history in a ledger, Drugledger can easily find out the drug transaction's origin in the network, which makes Drugledger more transparent among the stakeholders on the drug supply chain network.

The main challenge in traditional systems is to provide security for data. The proposed system uses smart contracts and blockchain to provide the security for data over transfer and storage and reduces the chances for entering counterfeit drugs into the supply chain, as well as preventing the data from attacks. By providing encrypted data, smart contracts preventing data loss and make data transfer much more secure. Blockchain in the proposed system holds the potential to enhance the supply chain's security, integrity, data provenance, and functionality. The performance comparison between the existing and the proposed system is detailed in [Table 10.1](#).

The proposed system facilitates all phases of the supply chain. The system enables streamlined visibility of movement and the stakeholders through which drugs transit in the supply chain. The improved traceability facilitates the optimization of flows of goods and an efficient stock

Table 10.1 Performance comparison of the existing system with the proposed system.

Performance measures	Existing system	Proposed system
Central dependency	Yes	No
Security	Low	Moderately high
Smart contract management	No	Ethereum smart contracts
Encryption	No	Public key cryptography
Error rate	Moderate	Low
User privacy	Low	High
Traceability	Low	High

management system. The system also provides decentralization to ensure each node's participation in the data flow. To reduce the error rate, the system focuses on data digitization and eliminates the paper trail used by doctors. Digital prescriptions make admittance of patients easier and simpler.

4. Blockchain use cases and its applications in the pharmaceutical industry

The blockchain is found to be the most useful, efficient, secure, and well-suited technology by all key partners in the pharmaceutical industry. They are actively engaged in exploring several real-world use cases involving tracking drugs in the drug supply chain, medical products provenance, and pharmaceutical supply chain governance rule and regulations ([Chilamkurti et al., 2021](#)). There are number of use cases of blockchain in pharmaceuticals discussed subsequently.

4.1 Prevention of counterfeit drugs and medical products

It is difficult to keep track of drugs or medical product status in the pharmaceutical supply chain where products move between many entities including manufacturers, distributors, hospitals, and patients; thereby the supply chain is prone to counterfeiters. Blockchain introduces provenance of data that avoids forgeries in the pharmaceutical supply chain. Blockchain's ability to provide complete traceability of drugs helps to identify the exact origin where the supply chain breaks down. US company Blockverify has introduced a project where a blockchain approach is used into the pharmaceutical supply chain to avoid forgeries. It uses the Bitcoin blockchain together with a private side chain where the private key of every product or drug is stored in a public blockchain. Every product is also given a tracking number to trace the change of ownership, and every change is stored in the private blockchain.

4.2 Compliance and governance in pharmaceutical supply chain

As supply chain drugs and medical products frequently move from one stakeholder to another stakeholder, it becomes necessary to adhere to important logistics rules and regulations by the stakeholders' companies that include guidelines for drug handling, transport, and storage. There are a number of operating constraints that need to be monitored, for example, maintaining temperature range for vaccines, humidity, air quality limits, and

other environmental conditions that directly influence quality and efficiency of medicines throughout the supply chain. Blockchain technology enables the consumer to easily check and monitor any event occurrence at any point of time. Smart contracts, programmed within the blockchain, can be executed automatically when compliance conditions are not adhered to by notifying the relevant stakeholders in the supply chain.

Properties like inherent transparency, immutability, decentralization, and distributed nature enable blockchain to add compliance measures and governance guidelines within the supply chain in a better way.

4.3 Improving clinical trial management

Blockchain integration into clinical trial management gives improved and more reliable clinical trial data. There are a number of use cases of blockchain that have significant potential in pharmaceuticals. Blockchain is a decentralized, immutable ledger that ensures transparency and recorded clinical trial data in a more secure, verified, and unfalsified manner. Blockchain prevents clinical trial results from tampering by suspicious or false stakeholders, thus improving the reliability of data. Blockchain can also be used to maximize the number of patients recruited for clinical experiments. In the blockchain system, patients can store and have access to and control of their personal medical records, as well as being able to make them visible to recruiters for clinical trials of their choice. A blockchain pharmaceutical startup called Exochain, based on blockchain, manages secure storage of patient medical data. It also allows individual patients to control the interaction with their private data from clinical trial researchers.

5. Open Challenges related to blockchain in the pharmaceutical industry

The blockchain approach has a number of potential applications in healthcare industries. However, there are many open research challenges and issues with the implementation and adoption of blockchain in the pharmaceutical industry. Following are some of the key challenges:

- **Security:** Blockchain systems have a number of safety vulnerabilities due to their structural design and working process. These blockchain protection vulnerabilities are associated with the traditional consensus system responsible for verifying and confirming transactions in a blockchain. The consensus system is incapable of handling security threats in the shared blockchain mechanism.

- **Privacy:** Using blockchain-based electronic health record (EHR) to provide a cryptographic system architecture to maintain confidentiality and reliability of individual patient medical data is one of the key challenges. Blockchain-based EHR takes a lot of computation power and time to complete all the tasks. Further, to add a new patient to the network requires a number of various verification steps even for a genuine patient.
- **Latency and throughput restrictions:** Bitcoin blockchain takes a lot of time to verify transactions (approximate transaction latency is 10 min) in the network, while the usual database methods take less time (a couple of seconds) to respond to a transaction.
- **Blockchain size:** When all nodes in a network conduct a transaction at same time, then the IoMT mechanism fails to handle the tiny size of blockchains.

Open challenges and issues related to blockchain in the pharmaceutical industry are shown in [Fig. 10.9](#).

- **Scalability:** The requirement for blockchain in the pharmaceutical industry is to manage and process a large number of health-related transactions, which decreases the performance of the system. Hence, blockchain size in the pharmaceutical industry directly affects the speed of the process, which means a bigger blockchain size will result in a

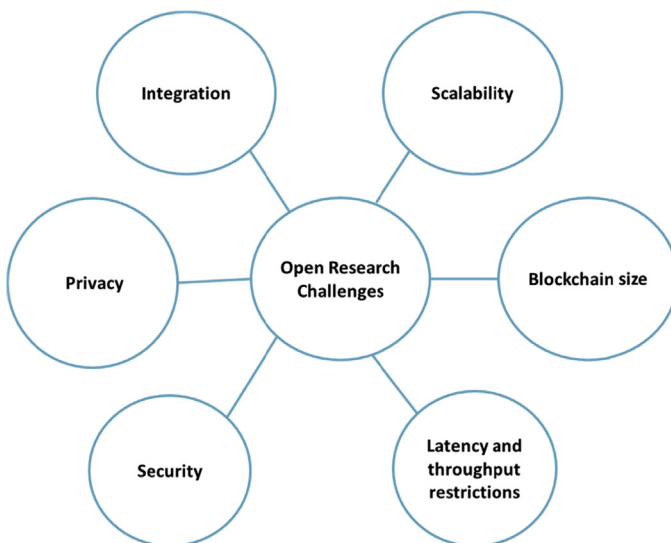


Figure 10.9 Challenges related to blockchain in pharmaceutical industries.

slower process. System scalability becomes a major challenge when the number of transactions performed by different entities increases (Kumar et al., 2021).

- **Integration:** In the pharmaceutical industry, blockchain-based solutions need to incorporate multiple distributed applications and platforms to operate, and this leads to security and interoperability issues.

6. Future direction

First, there is a need to see how the smallest amount of time can be achieved for a transaction to gain consensus. Further, there is also a requirement to check the possibility of reducing the cost of implementation, so the technology can be commercially appropriate for larger enterprise solutions. There also exists a significant need for more education and awareness for adoption of blockchain technology among the stakeholders of the pharmaceutical industry. The development of technical standards that concern the integration of blockchain into existing healthcare systems is also another area where researchers can see further solutions of blockchain technology. As a unifying solution, blockchain offers the potential for managing the supply chain of drugs. The decentralized architecture would promote real end-to-end supply management and encourage multiple stakeholders to participate, rather than being involved in this business at a greater level as unreliable, patchwork database approaches. Interoperability is the biggest problem involved in this project. Scalability, storage, social acceptance, and expectations need to be achieved. Blockchain primarily suffers from the problem of interoperability where, with sufficient need, the communication provider requires careful supply study. This poses many barriers to the success of data sharing.

In the future view of this technology, this industry produces enormous quantities of technology. Initiatives exist on a larger basis. For many sectors to step forward in future years, integrating blockchain with IoT will have a major effect. In addition, many pharmaceutical industry stakeholders are looking at blockchain as a great opportunity for their lives to evolve. The service providers are required on a large scale in the supply chain management for enhancing patient's life quality.

7. Conclusion

Despite the fact that blockchain technology is still in its early stages, already, there is a drastic progression of its application in day-to-day life, especially

in the pharmaceutical industry. For blockchain to achieve its complete potential and be used in healthcare, it must address a number of significant challenges, the most significant of which are the data access controls and scalability. Accessibility to a massive pool of anonymized healthcare data could be exploited. Customized drug development, rationalization of health insurance expenses, and advancement of public health policies are some of the most significant benefits that implementation of this technology in healthcare could provide. Furthermore, patients will have access to private health records, giving them control over what information professionals obtain and hold, under what conditions, and for how long. This kind of patient involvement, when combined with the availability of data about healthcare providers, could lead to a new era of healthcare. To summarize, the literature on blockchain applications in healthcare systems is still severely lacking. Health practitioners and decision-makers should be more attentive in terms of practicing medicines by implementing blockchain technologies in order to provide more security.

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